

L2 - 9.5. Atomic Bird Goes Purple #2

Task 1 - Introduction

Task	Base Tactics	Reference Techniques	Implemented Actions
#2	• TA003: Persistence • TA004: Privilege Escalation • TA005: Defense Evasion • TA006: Credential Access	• T1036.004 • T1552.001 • T1078.003	• Cleartext Data Search • Account Creation
#3	• TA003: Persistence • TA004: Privilege Escalation • TA007: Discovery • TA009: Collection • TA0040: Impact	• T1012 • T1112 • T1491 • T1543.003	• Service Creation • Defacement • Filetype Modification • Planting Reverse Shell in Registry

Task 2 - In-Between - Discover and Hide

Reference Techniques	Given atomic tests are inspired by the following techniques. • T1552.001 Unsecured Credentials: Credentials In Files • T1078.003 Valid Accounts: Local Accounts
Storyline	Purple Team aims to simulate discovering credentials in cleartext files and creating local accounts with a masquerading/typosquatting mindset. As a team member, your task is to discover the cleartext credentials, create accounts with given custom atomics and evaluate the generated artefacts.
Objective	Experiencing the potential impacts and artefacts of storing cleartext data, unprotected credentials and decoy accounts.

The planned tests for this case are listed below.

T0002-1 TASK-2.1 Search cleartext data

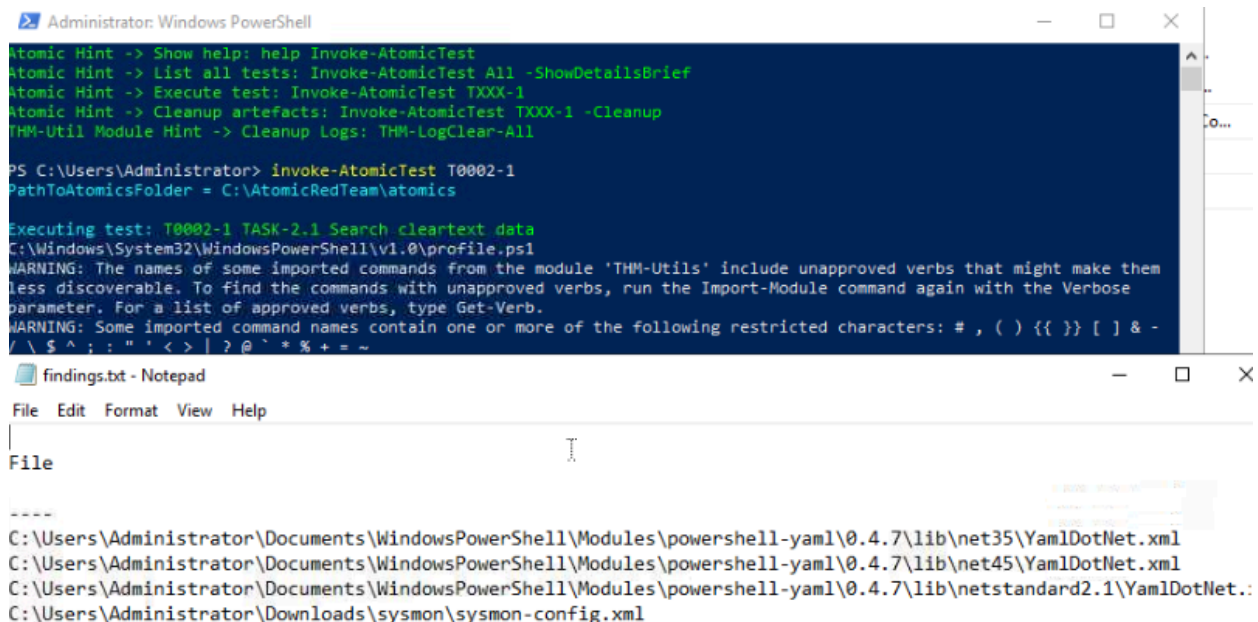
T0002-2 TASK-2.2 Create clone/decoy account

Note: You can revert the system modification and file change activities by using the cleanup command of the executed technique!

Answer the questions below

Execute test T0002-1 and open the document created on the Desktop. Which PowerShell library file is detected?

YamlDotNet.xml (After executing the Atomic Test, a TXT file was created on the desktop)



The screenshot shows two windows. The top window is 'Administrator: Windows PowerShell' with a blue background. It displays the following commands and output:

```
Atomic Hint -> Show help: help Invoke-AtomicTest
Atomic Hint -> List all tests: Invoke-AtomicTest All -ShowDetailsBrief
Atomic Hint -> Execute test: Invoke-AtomicTest TXXX-1
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup
THM-Util Module Hint -> Cleanup Logs: THM-LogClear-All

PS C:\Users\Administrator> invoke-AtomicTest T0002-1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T0002-1 TASK-2.1 Search cleartext data
C:\Windows\System32\WindowsPowerShell\v1.0\profile.ps1
WARNING: The names of some imported commands from the module 'THM-Utils' include unapproved verbs that might make them less discoverable. To find the commands with unapproved verbs, run the Import-Module command again with the Verbose parameter. For a list of approved verbs, type Get-Verb.
WARNING: Some imported command names contain one or more of the following restricted characters: # , ( ) { } [ ] & - / \ $ ^ ; : " ' < > | ? @ " * % + = ~

findings.txt - Notepad
```

The bottom window is 'findings.txt - Notepad' with a white background. It shows the following text:

```
-----
C:\Users\Administrator\Documents\WindowsPowerShell\Modules\powershell-yaml\0.4.7\lib\net35\YamlDotNet.xml
C:\Users\Administrator\Documents\WindowsPowerShell\Modules\powershell-yaml\0.4.7\lib\net45\YamlDotNet.xml
C:\Users\Administrator\Documents\WindowsPowerShell\Modules\powershell-yaml\0.4.7\lib\netstandard2.1\YamlDotNet.:
C:\Users\Administrator\Downloads\sysmon\sysmon-config.xml
```

Now go to the atomics path and update the executed script to include all "bak" files. What is the code snippet that needs to be added to the code? ,*.bak

(Went through the path that was written in the downloaded file)

Run the cleanup command for the test T0002-1 and re-execute the test. Open the output file, and investigate the detected files. What is the secret key?

L1LAFLHQ5peGsJh7Pee8wHFY1SBQHe85A1HZhVrK47Yf6cqmH3n8 (After cleaning up the logs, I ran Invoke-AtomicTest again and got a new TXT file.)

```
PS C:\Users\Administrator> invoke-AtomicTest T0002-1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T0002-1 TASK-2.1 Search cleartext data
C:\Windows\System32\WindowsPowerShell\v1.0\profile.ps1
WARNING: The names of some imported commands from the module 'THM-Utills' include unapproved verbs that might make them
less discoverable. To find the commands with unapproved verbs, run the Import-Module command again with the Verbose
parameter. For a list of approved verbs, type Get-Verb.
WARNING: Some imported command names contain one or more of the following restricted characters: # , ( ) { } [ ] & -
/ \ $ ^ ; : " ' < > | ? @ ` * % + = ~
Atomic Hint -> Show help: help Invoke-AtomicTest
Atomic Hint -> List all tests: Invoke-AtomicTest All -ShowDetailsBrief
Atomic Hint -> Execute test: Invoke-AtomicTest TXXX-1
Atomic Hint -> Cleanup artefacts: Invoke-AtomicTest TXXX-1 -Cleanup
THM-Util Module Hint -> Cleanup Logs: THM-LogClear-All
Done executing test: T0002-1 TASK-2.1 Search cleartext data
PS C:\Users\Administrator>
```

old.bak - Notepad

File Edit Format View Help

Backup!

Wallet: 1K

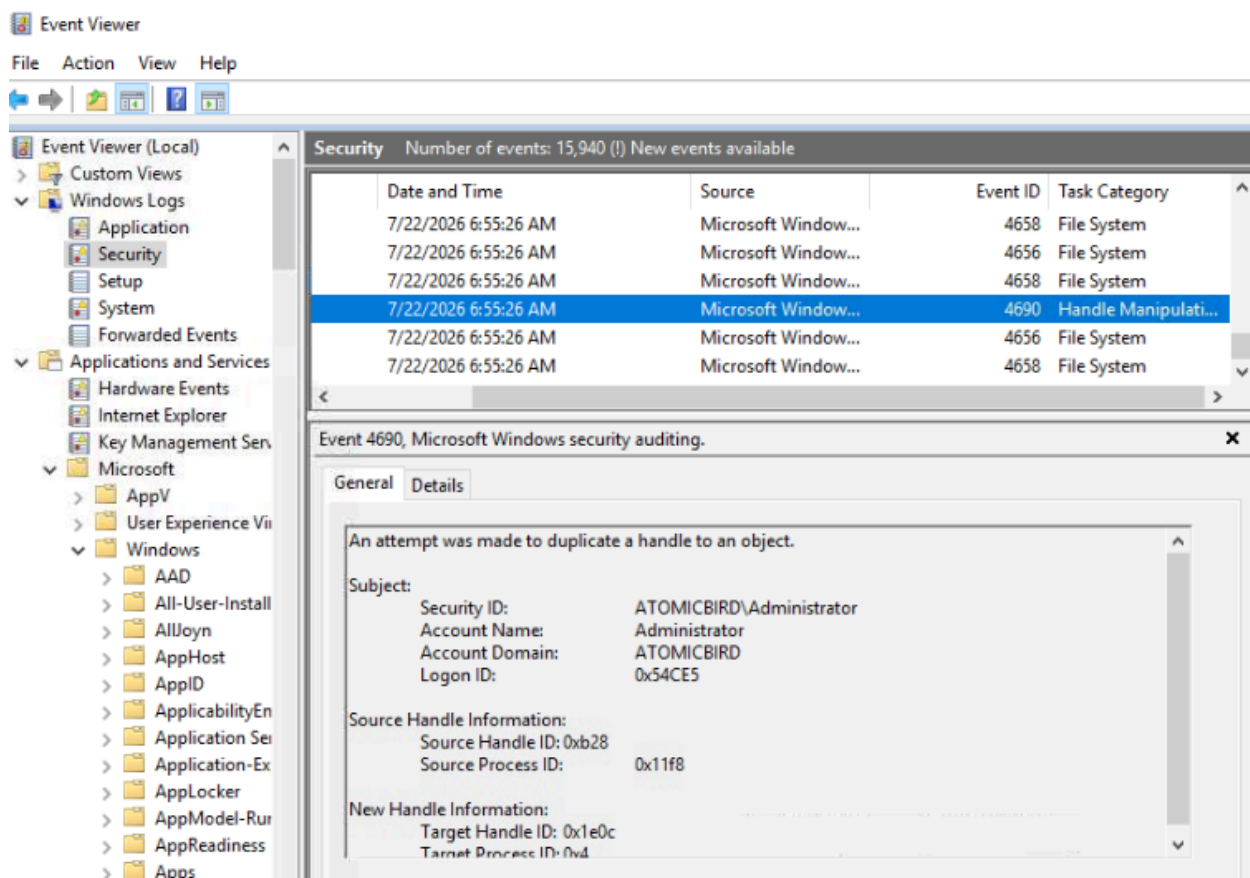
Secret: L1LAFHQ5peGsJh7Pee8wHFY1SBQHe85A1HZhVr

File

C:\Users\Administrator\Documents\WindowsPowerShell\Modules\poi
C:\Users\Administrator\Documents\WindowsPowerShell\Modules\poi
C:\Users\Administrator\Documents\WindowsPowerShell\Modules\poi
C:\Users\Administrator\Documents\WindowsPowerShell\inf.bak
C:\Users\Administrator\Downloads\sysmon\sysmon-config.xml
C:\Users\Administrator\old.bak

Execute test T0002-2 and investigate logs. What is the new account name?

Adminstrator



Task 3 - Manipulate, Deface, Persistence

Reference Techniques	Given atomic tests are inspired by the following techniques. • T1491 Internal Defacement • T1112 Modify Registry • T1543.003 Create or Modify System Process: Windows Service • T1012 Query Registry
Storyline	Purple Team aims to simulate creating custom services, manipulating file extensions, modifying the registry keys, and creating custom registry keys. As a team member, your task is to discover the cleartext credentials, create accounts with given custom atomics and evaluate the generated artefacts.
Objective	Experiencing the potential impacts and artefacts of creating services, file and registry modifications.

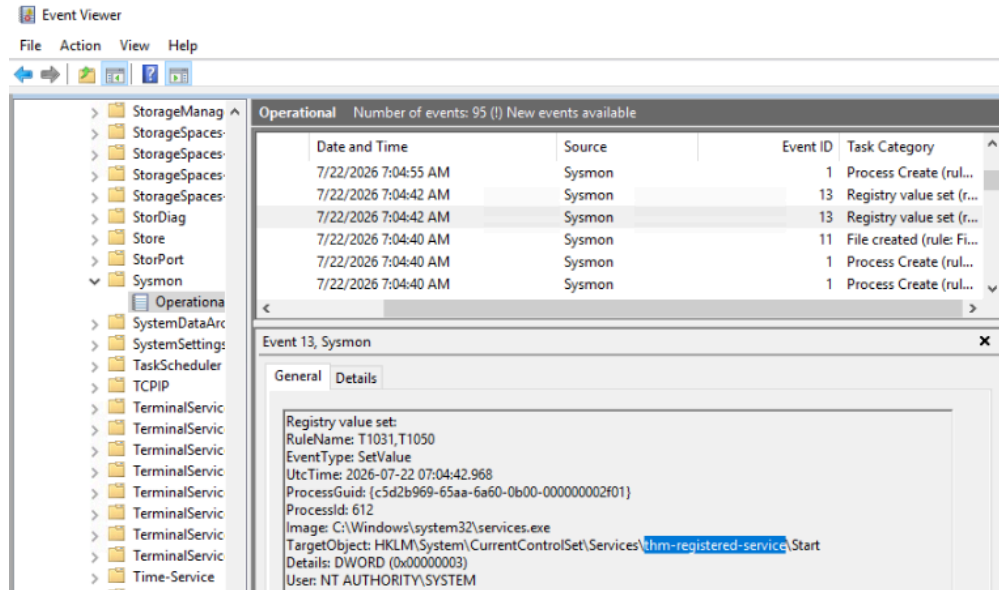
The planned tests for this case are listed below.

```
T0003-1 TASK-3.1 Internal service creation
T0003-2 TASK-3.2 Defacement with registry
T0003-3 TASK-3.3 File changes like a ransom
T0003-4 TASK-3.4 Planting reverse shell command in the registry
```

Note: You can revert the system modification and file change activities by using the cleanup command of the executed technique!

Answer the questions below

Execute test **T0003-1**. What is the name of the created service? thm-registered-service

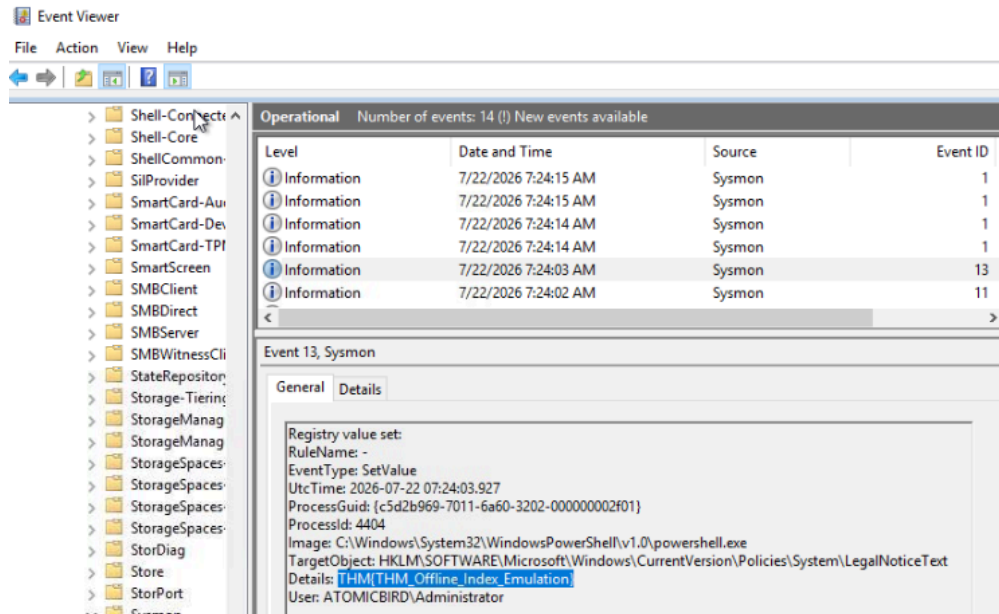


Which image is used to set the registry value for the created service?

C:\Windows\system32\services.exe

Execute test **T0003-2**. What is the ransom note?

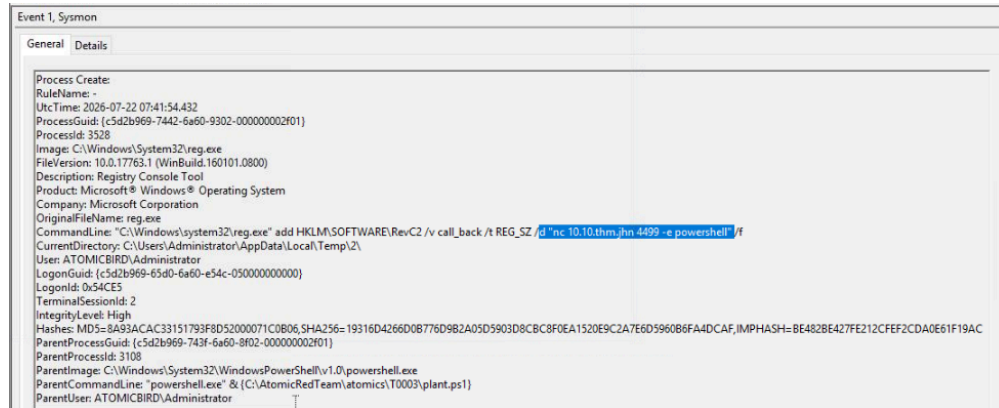
THM{THM_Offline_Index_Emulation}



Execute test **T0003-3**. What is the updated file extension?

.thm-jhn

Execute test T0003-4. What is the assigned value of the malicious registry value?



Task 4 - Conclusion

In this room, we covered the implementation of custom atomic tests and detection for Purple Team, Threat Emulation and Detection Engineering exercises.